

JONATHAN SANSO

Full-Stack / AI Engineer | LangChain/LangGraph, On-prem LLM & RAG, Face Recognition, OCR/NLP/STT | Multi-tenant SaaS (Supabase RLS)

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SUMMARY

Full-Stack Developer with **4 years** of experience building web products end-to-end (multi-tenant SaaS, internal platforms, AI services). Core expertise in **tenant isolation** (Supabase RLS/Policies), **POS/orders lifecycle**, **RBAC**, **AI engineering** (LLM orchestration with **LangChain/LangGraph**, offline **RAG**, face recognition embeddings, OCR/NLP/STT), **legacy DB migrations**, and **security-first architecture** (MFA, CSRF, HMAC/JWT). Daily stack: **Next.js**, **React**, **TypeScript**, Tailwind, **Supabase/PostgreSQL**, **NestJS**, Prisma, **Python** (FastAPI, Pandas, **InsightFace**). Experienced with **Docker**, **NGINX**, **AWS**, Terraform, **Redis/BullMQ**, **SendGrid**, Mercado Pago API.

Core Competencies: Full-Stack Development | Software Engineering | AI Engineering | LangChain | LangGraph | LLM Orchestration | Offline RAG | LLM Evaluation | On-premise / Air-gapped AI | Vector Search | Face Recognition | OCR | Speech-to-Text | Multi-tenant SaaS | RBAC | Row-Level Security | REST APIs | Microservices | CI/CD | Legacy DB Migration | ETL | Agile / Scrum | Code Review

TECHNICAL SKILLS

Frontend: Next.js, React, TypeScript, JavaScript (ES6+), Tailwind CSS, HTML5, CSS3, Zustand, TanStack Query, Zod, react-hook-form, @dnd-kit, ShadCN UI

Backend: NestJS, Node.js, Prisma, FastAPI (Python), Express, GraphQL, REST APIs / RESTful, OpenAPI / Swagger, Microservices

Databases: PostgreSQL (Supabase RLS, Triggers, RPC, Indexes), MySQL, MariaDB, MongoDB, SQL Server, iBase (legacy migration)

DevOps & Infra: Docker, NGINX, Debian/Linux, AWS (EC2, RDS, S3, CloudFront), Terraform, CI/CD, GitHub Actions, Vercel, Git/GitHub, PM2

AI & Data: LangChain (LCEL), LangGraph, Ollama / Qwen2.5, local embeddings, vector store on MariaDB, structured output (JSON Schema), Pydantic guardrails, golden-set evals, Gemini API, InsightFace (512-d face embeddings), Tesseract OCR, faster-whisper STT, NLP, Pandas, NumPy, jsPDF, ExcelJS

Integrations: SendGrid (transactional email), Mercado Pago, Redis/BullMQ, Webhooks (HMAC signed)

Testing & Methods: Unit Testing, Jest, integration tests, Agile / Scrum, Kanban, Code Review, Git Flow

Security: Auth0, MFA/2FA, CSRF, Helmet/CORS, HMAC-SHA256, JWT, Rate Limiting, Input Sanitization

PROFESSIONAL EXPERIENCE

Niztech -- AI & Automation Engineer (Full-Stack) - Aug 2026 - Present

- Automate **legacy back-office platforms built in 2010/2011** with **React + Express + SQLite** internal tools that cross results from independent systems, cutting manual **Data Entry** work.
- Integrate platforms owned by third parties through **API keys, webhooks and queues** (create/read/update records and forms) where no native integration exists.
- Built **Excel + PDF OCR pipelines** with **LLM APIs** for **overtime hours, budgets and invoices**, with structured validation and **human-in-the-loop** review before write-back.
- Shipped an **Excel engine dashboard** that ingests spreadsheets into one platform where they are normalized, linked and queried together.

Ocean Stack -- Full-Stack Developer / Software Engineer (Multi-tenant SaaS) - Dec 2025 - Jun 2026

Products: Niappa POS | Oceans HR (ATS) oceansstack.com

- Shipped **weekly product increments** in a structured delivery loop: **Figma** as source of truth, daily alignment with **CTO + PM**, iterative releases, and **UX/UI polishing** (Next.js, TypeScript, Tailwind).
- Implemented **tenant-safe isolation** and **role-based permissions** across modules so each business only sees and mutates its own data (**Supabase Postgres, RLS/Policies**).
- Delivered **operational back-office flows** (products, ingredients, stock, counts, suppliers) with **strong data consistency** guarantees via DB automation (**Postgres triggers, indexes, SQL/RPC**).
- Implemented **orders at scale: table sessions + split checks**, resilient **order lifecycle**, and safe **edit/void/restore** flows keeping stock accurate end-to-end (**RPC, transactional integrity, idempotent stock adjustments**).
- Built **reporting exports** for operations: **PDF/Excel** (sales + cash/audit), plus productivity-focused UI improvements (**i18n ES/EN**, search, sorting, confirmations).
- Built **Oceans HR ATS module** serving **hundreds of CVs/month per tenant** across multiple active clients: **Kanban pipeline with drag-and-drop**, vacancy management, candidate profiles with notes/comments, **recruitment funnel reports**, **RBAC with company-level roles**, full **ES/EN i18n** (Next.js 16, React 19, TanStack Query v5, Zod v4, @dnd-kit, Supabase RLS).
- Engineered **AI-Powered CV Matching Engine**: **LLM integration** (Gemini API with **3-model cascade fallback** for HA), **PDF text extraction**, **weighted scoring across 8 criteria** (skills, seniority, role, education, location, salary, language, industry), **structured JSON output**, plus full frontend UX with drag-and-drop upload, color-coded score cards, per-criteria progress bars, and **automated candidate creation pipeline** (Next.js App Router, Supabase RLS).
- Implemented **SendGrid transactional email pipeline** triggered by **candidate pipeline phase changes** (**automated and semi-automated flows** with recruiter approval step): **templated emails per stage**, **Supabase triggers + queued dispatch**, **idempotency keys**, and **delivery audit log per tenant**.

Argentine Federal Penitentiary Service -- Full-Stack Developer / AI Engineer / Software Engineer - Jan 2024 - Present

Private repos - *Size2i* internal platform (NestJS + Next.js + Python/FastAPI)

- Shipped an **on-prem NL intelligence chatbot** over a **90+ table legacy database: SQL-RAG** (deterministic parameterized retrieval, **zero hallucinated facts**, zero data egress) + **grounded 100% local LLM** (Ollama/Qwen2.5, CPU-only, air-gapped) orchestrated with **LangChain LCEL - structured output** (JSON Schema), **token streaming**, inline **[n] citations**, feature-flagged rollout; cut a >45s legacy self-join to **sub-second** (Next.js, NestJS, Python/FastAPI, Prisma, MariaDB).
- Built a **LangGraph** orchestration layer with **semantic validation**, **feedback-driven retries** (best-attempt strategy) and a **map-reduce pipeline with parallel fan-out** per entity; backed by a **labeled golden set** and **A/B/C evaluation** across LLM backends to pick the production default from measured data.
- Built a production **Python Face Matcher service: InsightFace buffalo_I** (ONNX CPU) **512-d embeddings**, in-memory NumPy **cosine top-K** index, **per-UID centroid** enrollment, persisted as BLOBs in MariaDB (**FastAPI**).
- Built an **AI-assisted document intelligence pipeline**: **Tesseract OCR** (spa+eng, binarization), **hybrid PDF extraction** (PyPDF2 + pdf2image/Poppler 300 DPI fallback, NDJSON streaming), and **faster-whisper ASR** (INT8 CPU), all with **human-in-the-loop review**.
- Migrated **~110 GB of legacy iBase8 data** to **MySQL/MariaDB** across **80+ modules** (relational + PDFs, scans, ZIPs, Word): **normalized schema**, **ETL** with chunked resumable streaming, checksum integrity, integrated into the new **NestJS + Next.js** stack.
- Automated **PDF/Excel/Word reporting** and **secure document verification** (**HMAC-SHA256**, **JWT**, time-limited access, watermarking); implemented **MFA/CSRF/rate limiting**, **Redis/BullMQ** queues and **containerized infra** for **170 concurrent users** (Docker, NGINX, PM2, Debian).

VirtuaState -- Frontend Developer - May 2022 - Dec 2023

- Built a responsive marketing/portfolio site for a VR/AR studio with **SEO** and performance optimizations (HTML5, CSS3, JavaScript, mobile-first).
Live: <https://www.virtuastate.net/>

EDUCATION

Higher Degree in Programming - Teclab - Aug 2022 - Aug 2024

- Software fundamentals and delivery practices: databases, web development, UX, cloud basics, project management.

Full-Stack Web Development - CoderHouse - Jan 2022 - Mar 2023

- Delivered multiple end-to-end projects: frontend and backend foundations, web app architecture (HTML5, EJS, CSS/SASS, React, Node.js, MongoDB, Express, GraphQL).
Live: <https://react-e-commerce-j-sanso.vercel.app/>